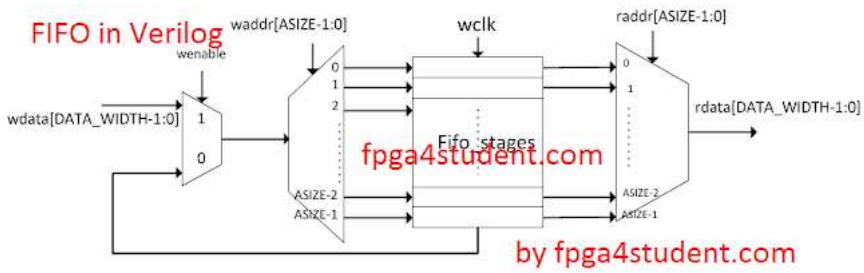
Verilog code for FIFO memory

In this project, Verilog code for FIFO memory is presented. The First-In-First-Out (FIFO) memory with the following specification is implemented in Verilog:

- **16 stages**
- **8-bit data width**
- **Status signals:**
  - **Full: high when FIFO is full else low.**
  - **Empty: high when FIFO is empty else low.**
  - **Overflow: high when FIFO is full and still writing data into FIFO, else low.**
  - **Underflow: high when FIFO is empty and still reading data from FIFO, else low.**
  - **Threshold: high when the number of data in FIFO is less than a specific threshold, else low.**



FIFO architecture

Verilog code for the FIFO memory:

```
// fpga4student.com: FPga projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for FIFO memory
// Top level Verilog code for FIFO Memory
module fifo_mem(data_out,fifo_full, fifo_empty, fifo_threshold, fifo_overflow, fif
  input wr, rd, clk, rst_n;
  input[7:0] data_in;   // FPGA projects using Verilog/ VHDL
  output[7:0] data_out;
  output fifo_full, fifo_empty, fifo_threshold, fifo_overflow, fifo_underflow;
  wire[4:0] wptr,rptr;
  wire fifo_we,fifo_rd;
  write_pointer top1(wptr,fifo_we,wr,fifo_full,clk,rst_n);
  read_pointer top2(rptr,fifo_rd,rd,fifo_empty,clk,rst_n);
  memory_array top3(data_out, data_in, clk,fifo_we, wptr,rptr);
  status_signal top4(fifo_full, fifo_empty, fifo_threshold, fifo_overflow, fifo_un
 endmodule
// fpga4student.com: FPga projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for FIFO memory
// Verilog code for Memory Array submodule
 module memory_array(data_out, data_in, clk,fifo_we, wptr,rptr);
  input[7:0] data_in;
  input clk,fifo_we;
  input[4:0] wptr,rptr;
  output[7:0] data_out;
  reg[7:0] data_out2[15:0];
  wire[7:0] data_out;
  always @(posedge clk)
  begin
   if(fifo_we)
      data_out2[wptr[3:0]] <=data_in ;
  end
  assign data_out = data_out2[rptr[3:0]];
```

```
 endmodule
// fpga4student.com: FPga projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for FIFO memory
// Verilog code for Read Pointer sub-module
 module read_pointer(rptr,fifo_rd,rd,fifo_empty,clk,rst_n);
  input rd,fifo_empty,clk,rst_n;
  output[4:0] rptr;
  output fifo_rd;
  reg[4:0] rptr;
  assign fifo_rd = (~fifo_empty)& rd;
  always @(posedge clk or negedge rst_n)
  begin
   if(~rst_n) rptr <= 5'b000000;
   else if(fifo_rd)
    rptr <= rptr + 5'b000001;
   else
    rptr <= rptr;
  end
 endmodule
// fpga4student.com: FPga projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for FIFO memory
// Verilog code for Status Signals sub-module
 module status_signal(fifo_full, fifo_empty, fifo_threshold, fifo_overflow, fifo_u
  input wr, rd, fifo_we, fifo_rd,clk,rst_n;
  input[4:0] wptr, rptr;
  output fifo_full, fifo_empty, fifo_threshold, fifo_overflow, fifo_underflow;
  wire fbit_comp, overflow_set, underflow_set;
  wire pointer_equal;
  wire[4:0] pointer_result;
  reg fifo_full, fifo_empty, fifo_threshold, fifo_overflow, fifo_underflow;
  assign fbit_comp = wptr[4] ^ rptr[4];
  assign pointer_equal = (wptr[3:0] - rptr[3:0]) ? 0:1;
  assign pointer_result = wptr[4:0] - rptr[4:0];
  assign overflow_set = fifo_full & wr;
  assign underflow_set = fifo_empty&rd;
  always @(*)
  begin
   fifo_full =fbit_comp & pointer_equal;
   fifo_empty = (~fbit_comp) & pointer_equal;
   fifo_threshold = (pointer_result[4]||pointer_result[3]) ? 1:0;
  end
  always @(posedge clk or negedge rst_n)
  begin
  if(~rst_n) fifo_overflow <=0;
  else if((overflow_set==1)&&(fifo_rd==0))
   fifo_overflow <=1;
   else if(fifo_rd)
    fifo_overflow <=0;
    else
     fifo_overflow <= fifo_overflow;
  end
  always @(posedge clk or negedge rst_n)
  begin
  if(~rst_n) fifo_underflow <=0;
  else if((underflow_set==1)&&(fifo_we==0))
   fifo_underflow <=1;
   else if(fifo_we)
    fifo_underflow <=0;
    else
     fifo_underflow <= fifo_underflow;
  end
 endmodule
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```

```
// Verilog project: Verilog code for FIFO memory
// Verilog code for Write Pointer sub-module
 module write_pointer(wptr,fifo_we,wr,fifo_full,clk,rst_n);
  input wr,fifo_full,clk,rst_n;
  output[4:0] wptr;
  output fifo_we;
  reg[4:0] wptr;
  assign fifo_we = (~fifo_full)&wr;
  always @(posedge clk or negedge rst_n)
  begin
   if(~rst_n) wptr <= 5'b000000;
   else if(fifo_we)
    wptr <= wptr + 5'b000001;
   else
    wptr <= wptr;
  end
 endmodule
```

Verilog testbench for the FIFO memory:

```
// 1. The timescale directive
 `timescale     10 ps/ 10 ps
// fpga4student.com: FPga projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for FIFO memory
// Verilog Testbench code for FIFO memory
 // 2. Preprocessor Directives
 `define          DELAY 10
 // 3. Include Statements
 //`include     "counter_define.h"
 module     tb_fifo_32;
 // 4. Parameter definitions
 parameter     ENDTIME      = 40000;
 // 5. DUT Input regs
 reg     clk;
 reg     rst_n;
 reg     wr;
 reg     rd;
 reg     [7:0] data_in;
 // 6. DUT Output wires
 wire     [7:0] data_out;
 wire     fifo_empty;
 wire     fifo_full;
 wire     fifo_threshold;
 wire     fifo_overflow;
 wire     fifo_underflow;
 integer i;
 // 7. DUT Instantiation
// fpga4student.com: FPga projects, Verilog projects, VHDL projects
 fifo_mem tb (/*AUTOARG*/
   // Outputs
   data_out, fifo_full, fifo_empty, fifo_threshold, fifo_overflow,
   fifo_underflow,
   // Inputs
   clk, rst_n, wr, rd, data_in
   );
 // 8. Initial Conditions
 initial
      begin
           clk     = 1'b0;
           rst_n     = 1'b0;
           wr     = 1'b0;
           rd     = 1'b0;
           data_in     = 8'd0;
      end
```

```
// 9. Generating Test Vectors
initial
     begin
          main;
     end
task main;
     fork
          clock_generator;
          reset_generator;
          operation_process;
          debug_fifo;
          endsimulation;
     join
endtask
task clock_generator;
     begin
          forever #`DELAY clk = !clk;
     end
endtask
task reset_generator;
     begin
          #(`DELAY*2)
          rst_n = 1'b1;
          # 7.9
          rst_n = 1'b0;
          # 7.09
          rst_n = 1'b1;
     end
endtask
task operation_process;
     begin
          for (i = 0; i < 17; i = i + 1) begin: WRE
               #(`DELAY*5)
               wr = 1'b1;
               data_in = data_in + 8'd1;
               #(`DELAY*2)
               wr = 1'b0;
          end
          #(`DELAY)
          for (i = 0; i < 17; i = i + 1) begin: RDE
               #(`DELAY*2)
               rd = 1'b1;
               #(`DELAY*2)
               rd = 1'b0;
          end
     end
endtask
// 10. Debug fifo
task debug_fifo;
     begin
          $display("-----------------------------------------------");
          $display("------------------   -----------------------");
          $display("----------- SIMULATION RESULT ----------------");
          $display("--------------       -------------------");
          $display("----------------     ---------------------");
          $display("-----------------------------------------------");
          $monitor("TIME = %d, wr = %b, rd = %b, data_in = %h",$time, wr, rd, dat
     end
endtask
// 11. Self-Checking
reg [5:0] waddr, raddr;
reg [7:0] mem[64:0];
always @ (posedge clk) begin
```

```
        if (~rst_n) begin
            waddr     <= 6'd0;
        end
        else if (wr) begin
            mem[waddr] <= data_in;
            waddr <= waddr + 1;
        end
        $display("TIME = %d, data_out = %d, mem = %d",$time, data_out,mem[raddr]);
        if (~rst_n) raddr     <= 6'd0;
        else if (rd & (~fifo_empty)) raddr <= raddr + 1;
        if (rd & (~fifo_empty)) begin
            if (mem[raddr]
             == data_out) begin
                $display("=== PASS ===== PASS ==== PASS ==== PASS ===");
                if (raddr == 16) $finish;
            end
            else begin
                $display ("=== FAIL ==== FAIL ==== FAIL ==== FAIL ===");
                $display("-------------- THE SIMUALTION FINISHED ------------");
                $finish;
            end
        end
    end
    //12. Determines the simulation limit
    task endsimulation;
        begin
            #ENDTIME
            $display("-------------- THE SIMUALTION FINISHED ------------");
            $finish;
        end
    endtask
    endmodule
```

The Verilog testbench is to debug and verify if the FIFO correctly operates. Besides, it is necessary to take a look at the simulation waveform and memory to see how data flows.

VHDL code for the FIFO Memory here

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